

PeakATail — Tunable Parameters & Filters

Complete reference generated from the code (RunConfig schema)

Contents

How to read this: which knobs move precision vs recall	2
The tuning map	2
Three things worth knowing before you tune	2
The measured trade-off (professor's benchmark, PBMC 10k)	2
ema run — pipeline parameters	3
1. Input & BAM reading	3
2. Peak calling — core	3
3. Peak calling — poly(A) evidence (clip_seeded)	4
4. Cleavage-site correction	6
5. PAS filters (internal priming / annotation / atlas)	7
6. Cell & matrix filters	9
7. Preprocessing & clustering	9
8. Performance & parallelism	11
9. Output, logging & plots	11
10. Other	12
ema switch — downstream analysis parameters	15
switch length	15
switch diff	17
switch trend	19
switch combine	22
switch match	23
switch geneview	24

How to read this: which knobs move precision vs recall

Every value in this document is extracted directly from the code (ema/cli/config_schema.py), so the defaults are exactly what the pipeline uses today. Set any of them as a CLI flag or as the YAML key under --config (the CLI always wins).

The tuning map

Goal	Turn these	Direction
Find more PAS (raise recall)	--lambda-fold-change (2.0), --polya-min-clip (6), --polya-min-purity (0.8), --polya-min-umis (1), --max-pas (5)	lower the thresholds / raise max-pas
Call fewer, cleaner PAS (raise precision)	--polya-mode filter (drops tier-2), --ip-filter-mode filter, --polya-min-umis 2, --annot-filter	enable / raise
Split more PAS inside one peak	--pas-gap (100), --min-pas-spacing (-1 = auto), --smoothing-window (50)	lower the gap / less smoothing
Coordinate accuracy (helps BOTH)	--auto-cleavage-offset	enable
Keep everything, just label it	--ip-filter-mode annotate, --polya-mode annotate, atlas annotate (all defaults)	default

Three things worth knowing before you tune

1. **--min-pas-prominence does nothing on the default strategy.** It is used by original / sierra_iterative only; the lambda strategies (lambda_poisson, lambda_gradient) ignore it. To loosen detection on lambda_gradient, use --lambda-fold-change.
2. **--max-pas 5 is a hard ceiling per peak.** A gene with more than five true PAS can never be fully recovered no matter what else you change.
3. **Filters only ever remove PAS.** They can raise precision but can never raise recall. Recall must be won at detection time (strategy + the thresholds in the first row above), or by relaxing the molecule gate --polya-min-umis.

The measured trade-off (professor's benchmark, PBMC 10k)

arm	Precision@100	Recall@100	F1@100
shipped lambda_gradient (pre-fix)	0.161	0.196	0.177
clip_seeded, both tiers	0.237	0.378	0.291
clip_seeded, tier-1 only	0.359	0.338	0.348 (best F1)
tier-1 + IP + $\geq$ \$2 molecules ("precision default")	~0.72	~0.17	~0.27

ema run — pipeline parameters

Every value below is **read directly from the code**, so the defaults are exact. Each row can be set as a CLI flag or as the YAML key in a `--config` file (CLI overrides YAML).

1. Input & BAM reading

CLI flag	YAML key	Default	Choices	What it does
<code>--bam-dir</code>	<code> `None` Single-BAM convenience. `--bam-files` </code>	None		Comma-separated multi-BAM list.
<code>--bam-threads</code>	<code>bam_threads</code>	4		pysam decompression threads per BAM.
<code>--barcode-tag</code>	<code>barcode_tag</code>	None		BAM tag holding the cell barcode (default CB).
<code>--cb-len</code>	<code>cb_len</code>	None		Cell-barcode length (bp).
<code>--config</code>	<code>' None YAML config; CLI flags override individual keys. -genome- fasta genome_fasta None Genome FASTA (indexed with pyfaidx/.fai) required by --ip-filter. -gtf gtf None'</code>		GTF annotation file.	

2. Peak calling — core

CLI flag	YAML key	Default	Choices	What it does
<code>--lambda-fold-change</code>	<code>lambda_fold_change</code>	2.0		Lambda fold-change cutoff for peak detection.
<code>--lambda-method</code>	<code>lambda_method</code>	"median"		Lambda estimator (median / mean / ...).
<code>--lambda-window</code>	<code>lambda_window</code>	5000		Background lambda estimation window (bp).

CLI flag	YAML key	Default	Choices	What it does
--max-pas	max_pas	5		Maximum PAS sites kept per peak.
--min-pas-prominence	min_pas_prominence	5.0		Static valley-depth threshold for the Tier-2 post-detection merger. Used by NON-lambda strategies (original, sierra_iterative). Lambda strategies (lambda_poisson, lambda_gradient) ignore
--min-pas-spacing	min_pas_spacing	-1		Hard distance floor (bp) for the post-detection PAS merger. Adjacent PAS within one peak whose gap is smaller than this are merged unconditionally (Tier 1). -1 (default)
--pas-gap	pas_gap	100		auto-detects median Minimum gap between PAS within a peak (bp).
--peak-strategy	peak_strategy	"lambda_gradient"		Peak-calling strategy (run -list-strategies to see).
--smoothing-window	smoothing_window	50		Coverage smoothing window (bp).
--tile-overlap	tile_overlap	10000		Tile overlap in bp.
--tile-size	tile_size	None		Tile size in bp (auto if unset).
--tiles	tiles	false		Enable tile-based peak calling (parallel pool).

3. Peak calling — poly(A) evidence (clip_seeded)

CLI flag	YAML key	Default	Choices	What it does
--polya-clip-filter	polya_clip_filter	"none"	none, f3844	Alignment filter on the poly(A) clip-evidence channel. 'none' (default) counts a molecule from every clip read read_check accepted — PCR duplicates cannot inflate it because duplicates share Tier-1 count window 'UP,DOWN' in bp, transcript orientation around a clip cluster's cleavage site (clip_seeded strategy only). A cluster that overlaps no coverage peak is counted from every
--polya-count-window	polya_count_window	"auto,25"		Collect read-level poly(A) soft-clip evidence during peak calling (default on). Annotate-only by default: each PAS's clip support – distinct (barcode, UMI) MOLECULES – is written into BED
--polya-evidence	polya_evidence	"on"	on, off	Minimum terminal soft-clip length AND minimum A/T run adjacent to the alignment boundary for a read to count as poly(A) evidence (default 6; measured 92x wrong-end specificity on PBMC).
--polya-min-clip	polya_min_clip	6		

CLI flag	YAML key	Default	Choices	What it does
<code>--polya-min-purity</code>	<code>polya_min_purity</code>	0.8		Minimum A (forward) / T (reverse) fraction over the clipped bases (default 0.8).
<code>--polya-min-umis</code>	<code>polya_min_umis</code>	1		Minimum distinct (CB, UMI) MOLECULES (reads without a UB tag count as one molecule each) for a clip cluster to be emitted as a tier-1 PAS, and the unit of BED column 5
<code>--polya-mode</code>	<code>polya_mode</code>	"annotate"	annotate, filter, require	(clip_seeded strategy) How poly(A) support is enforced (only relevant when <code>-polya-evidence</code> is on). 'annotate' (default) keeps every PAS. 'filter' drops PAS with zero clip support (BED score 0 == coverage-only tie)
<code>--polya-seed-window</code>	<code>polya_seed_window</code>	25		Single-linkage gap in bp for clustering clip sites into PAS candidates (clip_seeded strategy only; default 25).
<code>--polya-window</code>	<code>polya_window</code>	100		Half-window in bp around a PAS's strand-aware 3' base within which clip molecules count as support for that PAS (default 100 — the benchmark's matching cutoff).

4. Cleavage-site correction

CLI flag	YAML key	Default	Choices	What it does
--auto-cleavage-offset	auto_cleavage_offset	false		Data-driven 3' cleavage-offset estimation (issue #72). When set, the offset is inferred per run from the called peaks and -genome-fasta (genomic A-fraction crest + AATAAA density downstream
--cleavage-offset	cleavage_offset	0		3' cleavage-site offset correction (bp; issue #72). Called peak 3' ends stop ~90-105 nt short of the true cleavage site because 10x R2 coverage runs out before the poly(A) junction. When > 0

5. PAS filters (internal priming / annotation / atlas)

CLI flag	YAML key	Default	Choices	What it does
--annot-filter	annot_filter	false		Enable annotation filter: keeps only PAS overlapping an annotated gene region. Uses -annotation-bed if given, otherwise the GTF-derived gene BED (requires -gtf).
--annotation-bed	annotation_bed	None		Annotation BED for --annot-filter. Optional - defaults to the GTF-derived gene BED (directory_config.endbed) when unset.
--atlas	atlas	None		Reference PAS atlas BED.

CLI flag	YAML key	Default	Choices	What it does
--atlas-distance	atlas_distance	50	annotate, filter	Atlas snap distance (bp).
--atlas-mode	atlas_mode	"annotate"		How atlas snapping handles PAS with no atlas match (only relevant when --atlas is set). 'annotate' (default) KEEPS every PAS - non-matching PAS are alternative-PAS signal, not noise - and Include extended-3' annotations.
--include-extended	include_extended	false		Alternative internal-priming trigger: A/T fraction within the window.
--ip-a-fraction	ip_a_fraction	0.7		Minimum consecutive A's (or T's on - strand) flagged as internal priming.
--ip-a-stretch	ip_a_stretch	6	annotate, filter	Enable internal-priming filter: drops PAS near genomic A-rich stretches (requires --genome-fasta).
--ip-filter	ip_filter	false		Applied to the pos/neg PAS BEDs before gene assignment.
--ip-filter-mode	ip_filter_mode	"annotate"		How the internal-priming filter handles a flagged PAS (only relevant when --ip-filter is set). 'annotate' (default) KEEPS every PAS and records the internal_priming flag on the PAS ledger +
--ip-window-left	ip_window_left	10		Internal-priming check window, bp upstream of the PAS.

CLI flag	YAML key	Default	Choices	What it does
--ip-window-right	ip_window_right	30		Internal-priming check window, bp downstream of the PAS. Max distance for gene-end annotation (bp). 3'UTR length multiplier for extended-3' annotation.
--max-gene-distance	max_gene_distance	5000		
--utr-multiplier	utr_multiplier	2.0		

6. Cell & matrix filters

CLI flag	YAML key	Default	Choices	What it does
--min-cells	min_cells	3		Minimum cells expressing a PAS. Minimum cells (with non-zero counts for NB strategies) in each cluster group for a PAS to be included in differential testing. Default 10. Minimum PAS per cell (also bridges to filter_config.min_genes). Minimum reads per cell barcode.
--min-cells-per-group	min_cells_per_group	10		
--min-pas-per-cell	min_pas_per_cell	50		
--min-read	min_read	1500		

7. Preprocessing & clustering

CLI flag	YAML key	Default	Choices	What it does
--cluster-method	cluster_method	"leiden_tfidf"		Clustering strategy (leiden_tfidf / leiden_libsize / external).

CLI flag	YAML key	Default	Choices	What it does
--depth-corr-threshold	depth_corr_threshold	0.75		Pearson r threshold for removing LSI components correlated with sequencing depth (leiden_tfidf, ArchR-style). Default 0.75. Set to 1.0 to disable depth-correlation filtering.
--external-clusters	external_clusters	None		Path to external cluster labels TSV.
--marker-top-n	marker_top_n	200		Top-N markers per cluster for differential APA.
--match-method	match_method	"marker_overlap"		Cross-dataset cluster match strategy.
--n-neighbors	n_neighbors	30		Number of nearest neighbours for the kNN graph used by Leiden (leiden_tfidf default: 30; leiden_libsize default: 10 — set -n-neighbors explicitly to override).
--n-pcs	n_pcs	40		Number of principal components.
--n-svd-components	n_svd_components	50		Number of SVD/PCA components computed before filtering/neighbor graph (leiden_tfidf and leiden_libsize strategies). Default 50.
--n-top-hvg	n_top_hvg	2000		Number of highly variable genes/PAS selected before PCA (leiden_libsize strategy only). Default 2000.
--n-top-markers	n_top_markers	50		Number of top marker PAS per cluster.
--random-seed	random_seed	42		RNG seed for clustering reproducibility.

CLI flag	YAML key	Default	Choices	What it does
--resolution	resolution	1.0		Leiden resolution. Scale factor for Signac Method 1 TF-IDF (leiden_tfidf strategy). Default 10000. Adjust if your counts have very different dynamic range.
--tfidf-scale-factor	tfidf_scale_factor	10000.0		

8. Performance & parallelism

CLI flag	YAML key	Default	Choices	What it does
--peak-workers	peak_workers	None		Worker processes for per-chromosome peak calling (one job per contig and strand, merged deterministically). Default: auto, from the thread ceiling and free RAM. 1 selects the legacy single-p Max parallel workers (auto-detected if not set).
--threads	threads	None		

9. Output, logging & plots

CLI flag	YAML key	Default	Choices	What it does
--benchmark	benchmark	false		No-op (use scripts/validate_strategies.py). Don't write peakatail_.log next to outputs.
--log-level	`None` Logger level or `name=LEVEL` (comma-separated). `--no-log-file`	false		

CLI flag	YAML key	Default	Choices	What it does
--no-progress	`false` Suppress Rich progress bars. `--output` `output\_dir` `"emaout"` Output directory (timestamp suffix added automatically). `--quiet`	false		WARNING-and-up only.
--validate-db	validate_db	None		No-op (use scripts/validate_strategies.py).
--verbose	` 0`		Verbosity counter (-v / -vv).	

10. Other

CLI flag	YAML key	Default	Choices	What it does
' filenames None Optional dict of output filename overrides (YAML only – see RunConfig docstring for valid keys). -batch- size batch_size 10000 Worker batch size for streaming reads. -count- mode count_mode "cells" Unit the fisher strategy aggregates for its 2xN contingency table. 'cells' (default, D4, calibrated): each cell contributes at most once via per-cell PAS detection among gene-expressing cell -default- threshold default_threshold 5 Minimum read support per peak (peak-calling threshold). -dynamic- threshold dynamic_threshold false Use a per-window dynamic peak threshold. -fdr fdr 0.05 FDR threshold for differential APA calls. -floor- threshold floor_threshold 3 Minimum peak height (clamps dynamic threshold). -ignore- chro ignore_chro "MT,mt" Chromosomes excluded from	drop, gene	How to handle PAS that overlap no annotated UTR under iso- form_agg=per_isoform (switch length) or within_utr/between_utr (switch diff). 'gene' (default) KEEPS them via a gene-level fallba		

CLI flag	YAML key	Default	Choices	What it does
----------	----------	---------	---------	--------------

ema switch — downstream analysis parameters

switch length

Option	Default	Choices	Help
--list-strategies	flag		Print available length strategies and exit.
--threads	None		Max parallel workers (auto-detected if not set). Respected by ResourceManager as an absolute ceiling.
-v, --verbose	0		Increase verbosity. -v = DEBUG for ema.*; -vv = DEBUG everywhere.
-q, --quiet	flag		WARNING and up only. Overrides -verbose.
--log-level	None		Explicit logger level (DEBUG/INFO/WARNING/ERROR) or logger.name=LEVEL (repeatable: comma-separated).
--no-log-file	flag		Don't write peakatail_.log next to the outputs.
--no-progress	flag		Suppress Rich progress bars.
--config, -c	None		YAML config; CLI flags override individual keys.
--output, -o	"switch_out"		Output directory (timestamp suffix added automatically).
--plot-engine	"matplotlib"		Engines: 'matplotlib' (default), 'plotly', 'both', 'none', or comma list.
--plot-format	"all"		Restrict output formats. Default 'all' = png+svg+html as appropriate. Examples: 'svg' / 'png,svg' / 'html'.
--no-plots	flag		Disable all plotting (alias for -plot-engine none).
--h5ad, -i	Sentinel.UNSET		

Option	Default	Choices	Help
--pasbed	None		PAS BED6 (chrom/start/end/pas_id/score/strand). Supplies the strand + coordinates that define proximal vs distal. When omitted it is resolved by walking up from each -h AnnData layer holding raw counts. Default: prefer layers['counts'], else .X. Pass 'X' to force .X. Clustering overwrites .X with normalised values, so a clusters.h5ad wri Proceed even when the chosen matrix is non-integral (i.e. normalised, not counts). PDUI is then not a count ratio - diagnostics only, never a published number. Required when -isoform-agg=per_isoform.
--counts-layer	None		
--allow-non-count-matrix	flag		
--gtf	None		
--cluster-pairs	None		
--cluster-key	"leiden"		
--strategy, -s	"classic"		
--isoform-agg	"per_gene"	per_gene, per_isoform	Aggregation level — must match strategy vocabulary (per_gene collapses isoforms, per_isoform keeps them). How to collapse multiple isoforms when -isoform-agg=per_gene and the strategy tracks isoforms internally.
--isoform-collapse	"none"	none, mean, majority	Pseudocount added to counts before PDUI/entropy computation. Default 0.0 (original behaviour). Use 1.0 to avoid NaN on zero-count cells.
--pdui-pseudocount	0.0		

Option	Default	Choices	Help
--utr-unmatched	"gene"	drop, gene	How to handle PAS that overlap no annotated UTR under -isoform-agg=per_isoform. 'gene' (default) KEEPS them via a gene-level fallback (UTR-agnostic); 'drop' restores th

switch diff

Option	Default	Choices	Help
--list-strategies	flag		Print available diff strategies and exit.
--threads	None		Max parallel workers (auto-detected if not set). Respected by ResourceManager as an absolute ceiling.
-v, --verbose	0		Increase verbosity. -v = DEBUG for ema.*; -vv = DEBUG everywhere.
-q, --quiet	flag		WARNING and up only. Overrides -verbose.
--log-level	None		Explicit logger level (DEBUG/INFO/WARNING/ERROR) or logger.name=LEVEL (repeatable: comma-separated).
--no-log-file	flag		Don't write peakatail_.log next to the outputs.
--no-progress	flag		Suppress Rich progress bars.
--config, -c	None		YAML config; CLI flags override individual keys.
--output, -o	"switch_out"		Output directory (timestamp suffix added automatically).
--plot-engine	"matplotlib"		Engines: 'matplotlib' (default), 'plotly', 'both', 'none', or comma list.

Option	Default	Choices	Help
--plot-format	"all"		Restrict output formats. Default 'all' = png+svg+html as appropriate. Examples: 'svg' / 'png,svg' / 'html'.
--no-plots	flag		Disable all plotting (alias for --plot-engine none).
--h5ad, -i	Sentinel.UNSET		Per-dataset clusters.h5ad. Repeat for multi-dataset.
--pasbed	None		Optional PAS BED for context.
--counts-layer	None		AnnData layer holding raw counts. Default: prefer layers['counts'], else .X. Pass 'X' to force .X. The differential tests cast to integers, so testing normalised .X makes Proceed even when the chosen matrix is non-integral (i.e. normalised, not counts).
--allow-non-count-matrix	flag		Diagnostics only.
--gtf	None		Required when --isoform-agg=within_utr/between_utr (needed to resolve each PAS's 3'UTR isoform).
--cluster-pairs	None		c1, c2; c3, c4 — limit to specific pairwise contrasts.
--cluster-key	"leiden"		obs column defining the two (or more) groups to contrast – ANY column works, not just clusters: e.g. 'stage', 'celltype', 'leiden'. Combine with --cluster-pairs to pick
--isoform-agg	"per_gene"	per_gene, within_utr, between_utr, per_isoform	Scope of each differential test's background/denominator (applies to all strategies: fisher/nb_multi/nb_pairwise/mwu_per_cell). per_gene: each PAS vs the rest of its G

Option	Default	Choices	Help
--utr-unmatched	"gene"	drop, gene	How to handle PAS that overlap no annotated UTR under -isoform-aggr=within_utr/between_utr. 'gene' (default) KEEPS them via a gene-level fallback bucket (UTR-agnostic,
--marker-top-n	200		
--marker-method	"wilcoxon"		
--strategy, -s	"fisher"		Differential APA strategy (run -list-strategies to see).
--count-mode	"cells"	reads, cells	Aggregation unit for the fisher strategy's contingency table. 'cells' (default, D4): each cell counted at most once via per-cell PAS detection, de-pseudoreplicating the t
--fdr	0.05		
--per-worker-mb	300		
--min-cells-per-group	10		Minimum cells per group for a PAS to enter differential testing.
--log2fc-thresh	1.0		log2 fold-change threshold drawn on the volcano plot. Default 1.0.

switch trend

Option	Default	Choices	Help
--threads	None		Max parallel workers (auto-detected if not set). Respected by ResourceManager as an absolute ceiling.
-v, --verbose	0		Increase verbosity. -v = DEBUG for ema.*; -vv = DEBUG everywhere.
-q, --quiet	flag		WARNING and up only. Overrides -verbose.

Option	Default	Choices	Help
--log-level	None		Explicit logger level (DEBUG/INFO/WARNING/ERROR) or logger.name=LEVEL (repeatable: comma-separated).
--no-log-file	flag		Don't write peakatail_.log next to the outputs.
--no-progress	flag		Suppress Rich progress bars.
--config, -c	None		YAML config; CLI flags override individual keys.
--output, -o	"emaout"		Output directory (timestamp suffix added automatically).
--plot-engine	"matplotlib"		Engines: 'matplotlib' (default), 'plotly', 'both', 'none', or comma list.
--plot-format	"all"		Restrict output formats. Default 'all' = png+svg+html as appropriate. Examples: 'svg' / 'png,svg' / 'html'.
--no-plots	flag		Disable all plotting (alias for --plot-engine none).
--pdui	None		Input long TSV from ema switch length (has stage + value columns). Required for the default pdui metric unless --length-dir is given; also usable as a generic 'the input'
--metric	None	pdui, proportion, entropy	Convenience metric selector. Sets the value column (and, with --length-dir, the source TSV) to sane defaults: pdui -> 'pdui' on the classic table; entropy -> 'normalized\
--length-dir	None		A length/<celltype>/ directory from ema switch length (containing classic/proportion/shannon subdirs); used to auto-locate a metric's source TSV when its explicit -p

Option	Default	Choices	Help
<code>--proportion-tsv</code>	None		Per-PAS proportion TSV from ema switch length -s proportion (gene_id/cell/rank/proportion/stage columns). Used by -metric proportion and by -combine.
<code>--entropy-tsv</code>	None		Shannon entropy TSV from ema switch length -s shannon (gene_id/cell/normalized_entropy/stage columns). Used by -metric entropy and by -combine.
<code>--combine</code>	None		Comma list of metrics to combine into a consensus trend, e.g. 'pdui,proportion,entropy'. Each metric's source is located the same way as -metric (its —tsv or -
<code>--combine-min-agree</code>	None		Minimum number of VOTING metrics (see -include-entropy-in-vote) that must agree on increasing/decreasing for a gene's consensus to be called 'shortening'/'lengthening';
<code>--include-entropy-in-vote</code>	flag		Count entropy's direction toward the shortening/lengthening consensus vote. Off by default: rising entropy means more even PAS usage, not cleanly 'lengthening' the way a
<code>--stage-order</code>	Sentinel.UNSET		Comma- separated ordered stage labels, e.g. 'Normal,StageI,StageII,StageIII,StageIV,M
<code>--stage-col</code>	"cluster"		Column holding the stage/cluster label.
<code>--value-col</code>	"pdui"		Numeric value column (pdui/proportion/entropy); ignored for -metric/-combine proportion, which is derived.

Option	Default	Choices	Help
--gene-col	"gene_id"		Gene column for the per-gene trend table (skipped if absent).

switch combine

Option	Default	Choices	Help
--threads	None		Max parallel workers (auto-detected if not set). Respected by ResourceManager as an absolute ceiling.
-v, --verbose	0		Increase verbosity. -v = DEBUG for ema.*; -vv = DEBUG everywhere.
-q, --quiet	flag		WARNING and up only. Overrides -verbose.
--log-level	None		Explicit logger level (DEBUG/INFO/WARNING/ERROR) or logger.name=LEVEL (repeatable: comma-separated).
--no-log-file	flag		Don't write peakatail_.log next to the outputs.
--no-progress	flag		Suppress Rich progress bars.
--config, -c	None		YAML config; CLI flags override individual keys.
--output, -o	"switch_combined"		Output directory (timestamp suffix added automatically).
--plot-engine	"matplotlib"		Engines: 'matplotlib' (default), 'plotly', 'both', 'none', or comma list.
--plot-format	"all"		Restrict output formats. Default 'all' = png+svg+html as appropriate. Examples: 'svg' / 'png,svg' / 'html'.
--no-plots	flag		Disable all plotting (alias for -plot-engine none).
--input, -i	Sentinel.UNSET		LABEL=PATH for each stage-labelled clusters.h5ad (repeatable, >=2 required).

Option	Default	Choices	Help
--group-key	"stage"		obs column to stamp with the input LABEL.
--split-key	None		Optional obs column (e.g. 'celltype') to split by before combining.
--min-cells	1		Drop a group value with fewer than this many cells.

switch match

Option	Default	Choices	Help
--list-strategies	flag		Print available match strategies and exit.
--threads	None		Max parallel workers (auto-detected if not set). Respected by ResourceManager as an absolute ceiling.
-v, --verbose	0		Increase verbosity. -v = DEBUG for ema.*; -vv = DEBUG everywhere.
-q, --quiet	flag		WARNING and up only. Overrides -verbose.
--log-level	None		Explicit logger level (DEBUG/INFO/WARNING/ERROR) or logger.name=LEVEL (repeatable: comma-separated).
--no-log-file	flag		Don't write peakatail_.log next to the outputs.
--no-progress	flag		Suppress Rich progress bars.
--config, -c	None		YAML config; CLI flags override individual keys.
--output, -o	"switch_out"		Output directory (timestamp suffix added automatically).
--plot-engine	"matplotlib"		Engines: 'matplotlib' (default), 'plotly', 'both', 'none', or comma list.

Option	Default	Choices	Help
--plot-format	"all"		Restrict output formats. Default 'all' = png+svg+html as appropriate. Examples: 'svg' / 'png,svg' / 'html'.
--no-plots	flag		Disable all plotting (alias for --plot-engine none).
--h5ad, -i	Sentinel.UNSET		Per-dataset clusters.h5ad files.
--strategy, -s	"marker_overlap"		
--n-top-markers	50		
--mnn-components	30		SVD components for MNN shared embedding (--strategy mnn). Default 30.
--mnn-k-neighbors	10		Nearest neighbours for MNN search (--strategy mnn). Default 10.

switch geneview

Option	Default	Choices	Help
--threads	None		Max parallel workers (auto-detected if not set). Respected by ResourceManager as an absolute ceiling.
-v, --verbose	0		Increase verbosity. -v = DEBUG for ema.*; -vv = DEBUG everywhere.
-q, --quiet	flag		WARNING and up only. Overrides --verbose.
--log-level	None		Explicit logger level (DEBUG/INFO/WARNING/ERROR) or logger.name=LEVEL (repeatable: comma-separated).
--no-log-file	flag		Don't write peakatail_.log next to the outputs.
--no-progress	flag		Suppress Rich progress bars.
--config, -c	None		YAML config; CLI flags override individual keys.

Option	Default	Choices	Help
--output, -o	"switch_geneview"		Output directory (timestamp suffix added automatically).
--plot-engine	"matplotlib"		Engines: 'matplotlib' (default), 'plotly', 'both', 'none', or comma list.
--plot-format	"all"		Restrict output formats. Default 'all' = png+svg+html as appropriate. Examples: 'svg' / 'png,svg' / 'html'.
--no-plots	flag		Disable all plotting (alias for --plot-engine none).
--diff-tsv	Sentinel.UNSET		Differential TSV from ema switch diff (repeatable). Used to auto-select top-N genes by volcano score. Optional when --gene-id is supplied.
--pasbed	Sentinel.UNSET		pasbed.bed — PAS coordinates (BED6 format).
--h5ad, -i	Sentinel.UNSET		clusters.h5ad with leiden cluster labels and gene_id in var.
--gtf	None		Ensembl GTF for isoform structure (optional).
--top-genes	10		Number of top genes to auto-pick from --diff-tsv results.
--gene-id	Sentinel.UNSET		Explicit gene ID(s) to render (repeatable). Union-ed with --top-genes list.
--cluster-key	"leiden"		obs column carrying cluster labels.
--color-key	None		obs column carrying a condition per track (e.g. healthy / primary tumour / metastasis). Tracks are tinted by it and a legend is drawn, so the grouping is readable without

Option	Default	Choices	Help
<code>--subtitle</code>	<code>""</code>		Line shown under the gene title, e.g. the cell type the panel is restricted to. Naming it once in the header beats repeating it on every track label.
<code>--pas-distance-table</code>	<code>flag</code>		Draw a table of PAS coordinates and the distance between adjacent PAS beneath each panel, and write the full table next to the figure as <code>gene__pas_distances.csv</code> .
<code>--isoform-map</code>	<code>None</code>		<code>pdui_classic.tsv</code> from <code>switch length --isoform-aggregate --isoform-map</code> . Used to show which transcript's 3'UTR each PAS was assigned to in the distance table. The assignment is